

ERVA SENGUL

Computer Engineer · Artificial Intelligence & Machine Learning

Istanbul, Türkiye | +90 534 976 3877 | ervasengul4@gmail.com | linkedin.com/in/erva-sengul-016945237 | github.com/ervasengul

Computer engineer focused on applied artificial intelligence, with hands-on experience in LLM fine-tuning, NLP, and computer vision, and a published research paper. I enjoy building machine learning systems and deploying them reliably, from fine-tuning generative models to running them on cloud infrastructure. Having studied in the United States at a young age, I am a native-level English speaker comfortable in international environments. I am looking for a role where I can grow across machine learning, generative AI, and the infrastructure that runs it.

EDUCATION

Istanbul Medipol University

B.S., Computer Engineering

Istanbul, Türkiye

Oct 2022 - Jun 2026

- Completed a graduation project, Intelligent Cluster Optimizer - applying machine learning to infrastructure through ML-driven resource forecasting and optimization; published as a peer-reviewable research paper (see Publications).

Relevant Coursework: Introduction to Deep Learning, DevOps, Data Communication and Computer Networks, Algorithm Analysis.

Istanbul Koy Hizmetleri Anatolian High School

High School Diploma

Istanbul, Türkiye

Oct 2018 - Jun 2022

PUBLICATIONS

A. Karakaya, E. Sengul, and A. Kaplan. "Safety-Gated Autoscaling: A Multi-Layered Defense Architecture for Kubernetes Vertical Resource Optimization." arXiv:2607.26503, 2026. arxiv.org/abs/2607.26503

EXPERIENCE

TEKNOFEST 2025

Engineer Participant

Istanbul, Türkiye

Sep 2025

- Presented a real-time emotion recognition AI application built with Python, TensorFlow, Keras, and OpenCV.
- Developed a CNN pipeline for seven-class facial expression recognition using CK+ data augmentation and live webcam inference.

Medipol Healthcare Group

Software Engineer Intern

Istanbul, Türkiye

Aug 2025 - Oct 2025

- Supported IPAM modules used in international patient operations and hospital information workflows, handling secure patient data with high operational reliability.
- Visited the group's London office and saw the international patient systems in use there, gaining first-hand insight into cross-border healthcare operations.

TEI - TUSAS Engine Industries, Inc.

Student Participant

Istanbul, Türkiye

Mar 2025 - May 2025

- Completed TEI's competitive professional development program; analyzed engineering case studies across reliability, performance, and technical problem solving.

Conovel Information Technologies

Software Engineer Intern

Istanbul, Türkiye

Sep 2024 - Oct 2024

- Built full-stack features in ASP.NET Core MVC using C#, HTML, CSS, and SQL, with database-backed functionality to improve usability and performance.

PROJECTS

AI Pre-Evaluation System for Medical Tourism | Generative AI / NLP

- Fine-tuned MedGemma-4B with QLoRA on a clinical dataset to convert multilingual medical documents into structured JSON.
- Built an NLLB-200 translation pipeline with overlapping-window chunking and generated a deterministic clinical timeline from the extracted data.

Real-Time Emotion Recognition (TEKNOFEST 2025) | Computer Vision / Deep Learning

- Trained a CNN on the CK+ facial-expression dataset with data augmentation for seven-class emotion recognition, combined with SVM, LBP, and ORB features for real-time webcam inference.
- Built with Python, TensorFlow, Keras, and OpenCV; managed data preprocessing, model training, and real-time deployment.

Hospital Software Platform | In Development

- Currently developing a software product for hospital operations.

Intelligent Cluster Optimizer | *Graduation Project - Published (arXiv:2607.26503) - ML for Infrastructure*

- Applied machine learning to infrastructure: a Go-based Kubernetes optimizer using time-series forecasting for ML-driven resource right-sizing, achieving 20-40% cost savings on Google Kubernetes Engine.
- Designed a four-layer safety framework with memory-leak gating; validated with 1118 automated tests at 80.3% coverage.

Weather Forecast Modeling Using Interpolation | *Data Modeling*

- Built a forecasting workflow on NOAA data with Python preprocessing, interpolation, and time-series analysis for seasonal pattern detection.

TECHNICAL SKILLS

AI / Machine Learning: PyTorch, TensorFlow, Keras, Hugging Face, scikit-learn, OpenCV, LLM fine-tuning (QLoRA/LoRA), Generative AI, NLP, Computer Vision

LLMs & AI Tools: Claude, ChatGPT, Gemini, Claude Code

Languages: Python, Go, C#, C, SQL, JavaScript

MLOps & Cloud: Kubernetes, GKE, Docker, Git, Linux, CI/CD, Prometheus

Data & Web: SQL, Django, ASP.NET Core MVC